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Electrochemical synthesis of sulfonated benzothiophenes using 2-alkynylthioanisoles and sodium sulfinates†

Ming-Ming Zhang,[a] Yu Sun,[a] Wan-Wan Wang,[c] Kang-Kang Chen,[a] Wen-Chao Yang *[a] and Lei Wang *[b]

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Electrochemical sulfonylation/cyclization of 2-alkynylthioanisoles with sodium sulfinates was developed under catalyst-, external oxidant- and metal-free conditions. The electrosynthesis provides sustainable and efficient access to 3-sulfonated benzothiophenes with good substrate scope and functional group tolerance. This cascade radical process has been triggered through a sulfonyl radical addition to alkynes using sodium sulfinates under electrochemical conditions.

Radical cascade cyclization of internal alkynes, a powerful method for the synthesis of cyclic molecules by the introduction of different functional groups at both ends of the alkynes in one step, has attracted increasing attention.[1–3] From the viewpoint of synthetic chemistry, these radical cascade reactions can provide very convenient and efficient synthetic methods to obtain high value-added, complex molecular scaffolds with abundant bioactivities using relatively inexpensive alkynes. For example, Song and co-workers developed efficient and valuable methods to access 3-sulfonated benzothiophenes using photoredox-catalyzed or TBHP-promoted cascade cyclization of 2-alkynylthioanisoles.[2c,d] Recently, another novel cascade approach for the construction of 3-sulfonated benzothiophenes *via* a radical relay strategy was reported by Wu's group.[2e] Although significant progress has been made in recent years, most of the traditional transformations usually utilize transition-metal catalysts or stoichiometric amounts of oxidants. As a result, the development of a practical and green protocol to realize the radical cyclization of alkynes is still an urgent and attractive topic.

In this regard, electrocatalysis offers an ideal alternative solution since this reaction model utilizes electrons as traceless "reagents" to selectively activate small organic molecules, generating radical intermediates without the application of metal catalysts or dangerous chemical oxidants.[4] Over the past five years, the radical cyclization of internal alkynes *via* electrochemical means has blossomed in an exponential manner.[5] For example, Xu and other research groups have succeeded in the electrochemical cascade cyclization of monoalkynes or polyalkynes for the synthesis of N-heterocyclic products through the N-centered radical addition to a carbon–carbon triple bond with the formation of a vinyl radical intermediate.[6] Moreover, Lei and co-workers reported an electrooxidative radical cascade cyclization of ynones with sulfinic acids for the construction of sulfonated indenones, which obviates the use of oxidants and transition-metal catalysts.[7] Recently, another novel and efficient electrochemical intermolecular cascade [4 + 2] cyclization of alkynes with 1,3-dicarbonyl compounds was also developed by Tang and Pan's group for the synthesis of 1-naphthols.[8]

Mechanistically, the aforementioned radical cyclization of internal alkynes shares a similar pathway in the addition of a vinyl radical intermediate to a benzene ring (Scheme 1a). However, the utilization of a heteroatom as a vinyl radical acceptor has been less studied, probably due to the lack of easy-to-leave protecting groups on heteroatoms or a system that easily isomerizes after a vinyl radical attack. Only one case was achieved for the synthesis of imidazo-fused N-heteroaromatic compounds in 2018 by Xu's group, in which the nitrogen atom showed a strong preference for a vinyl radical acceptor instead of the carbon atoms in N-containing π-systems (Scheme 1b).[6c] As such, the exploration of novel and efficient approaches for the electrochemical cyclization of alkynes that apply heteroatoms as vinyl radical acceptors continues to be an important and challenging task in organic synthesis.

On the other hand, the synthesis of sulfone-containing derivatives with a myriad of biological activities is a quest in organic chemistry,[9] and the renaissance of electrochemical

[a]Institute of Pesticide, School of Horticulture and Plant Protection, Yangzhou University, Yangzhou 225009, P. R. China. E-mail: wccyang@126.com
[b]Advanced Research Institute and Department of Chemistry, Taizhou University, Taizhou, Zhejiang, 318000, P. R. China. E-mail: leiwang@chnu.edu.cn
[c]College of Pharmacy and Chemistry & Chemical Engineering, Taizhou University, Taizhou, Jiangsu 225300, P. R. China

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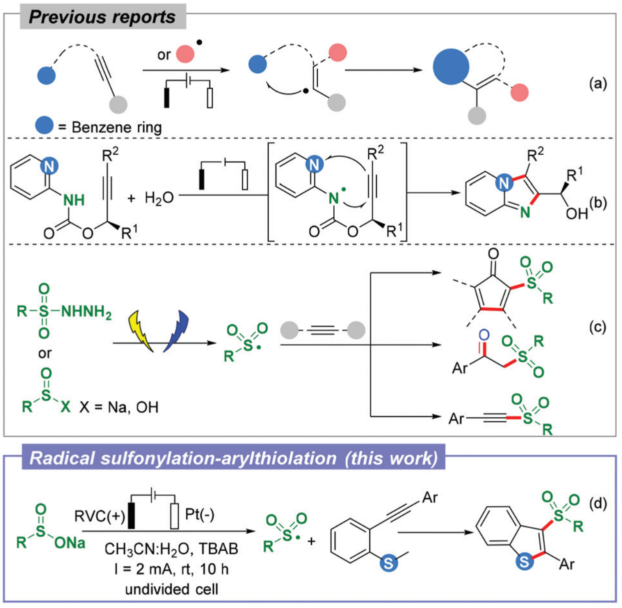
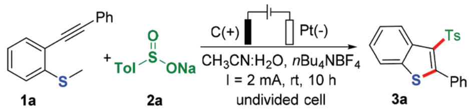


Scheme 1 The electrochemical radical cyclization or sulfonylation of alkynes.

Table 1 Optimization of the reaction conditions[a]



Entry	Variation from standard conditions	Yield[b] (%)
1	None	45
2	DMF instead of $CH_3CN:H_2O$	40
3	EtOH instead of $CH_3CN:H_2O$	35
4	$EtOH:H_2O$ instead of $CH_3CN:H_2O$	21
5	TBAB instead of n-Bu_4NBF_4	62
6	n-Bu_4NPF_6 instead of n-Bu_4NBF_4	35
7	NH_4Br instead of n-Bu_4NBF_4	37
8	$LiClO_4$ instead of n-Bu_4NBF_4	Trace
9[c]	10 mA instead of 2 mA	Trace
10[c,d]	RVC/Pt instead of C/Pt	77
11[c]	Pt/Pt instead of C/Pt	40
12[c]	C/Ni instead of C/Pt	34
13[c]	No electricity	n.d.

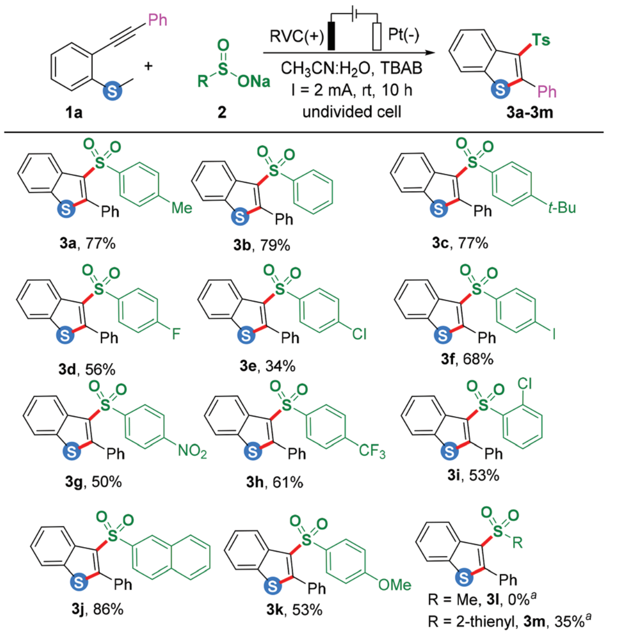
[a] Reaction conditions: 2-Alkynylthioanisole (**1a**, 0.20 mmol), sodium sulfinate (**2a**, 0.40 mmol), constant current = 2.0 mA, n-Bu_4NBF_4 (0.20 M), $CH_3CN:H_2O$ (1 : 1, 2.0 mL), rt, 10 h. [b] Isolated yield, n.d. = not detected. [c] TBAB instead of n-Bu_4NBF_4. [d] Faradaic efficiency: 41%.

synthesis has furnished unprecedented opportunities in the cascade sulfonylation of C–C unsaturated bonds.[10] Recently, several electrochemically induced radical functionalizations of alkynes with sulfonyl hydrazides or sulfinic (acid) sodium for the synthesis of SO_2-containing molecules have been reported (Scheme 1c).[7,11] The development in this field is not only a way of generating sulfone radicals, but also a means to design reaction substrates. Inspired by the recent developments in the area of electrochemical sulfonylation and our continuous research interest in the cascade reactions of alkynes and radical transformations,[12] herein we explore the possibility of sulfonylation of alkynes for the synthesis of a 3-sulfonated benzothiophene motif under mild electrochemical anodic oxidation conditions where the sulfur atom acts as the vinyl radical acceptor (Scheme 1d).[13]

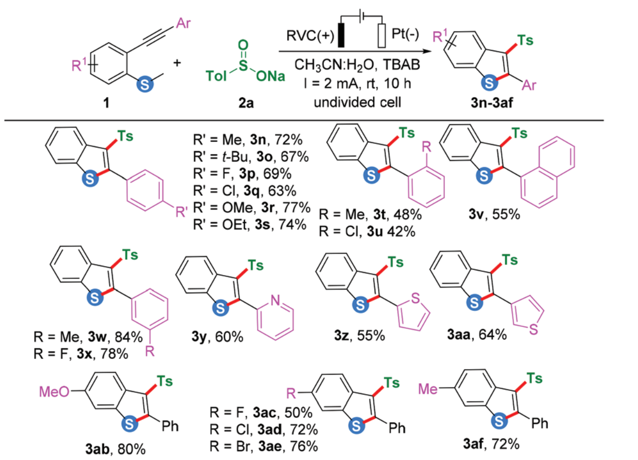
Initially, 2-alkynylthioanisole **1a** and sodium *p*-toluenesulfinate **2a** were chosen as the model substrates to investigate the conditions of electrochemical cyclization of alkynes that apply heteroatoms as vinyl radical acceptors in an undivided cell, as shown in Table 1. After a series of optimizations, the cascade product **3a** could be isolated in 45% yield under a constant current electrolysis at 2 mA in an undivided cell using n-Bu_4NBF_4 as the electrolyte in a mixture of $EtOH:H_2O$ (v/v 1 : 1) at room temperature (Table 1, entry 1). After that, we examined the effect of solvents, and our preliminary experimentation revealed that other solvents resulted in lower yields of **3a** (Table 1, entries 2–4). Among a series of electrolytes (Table 1, entries 5–8), the n-Bu_4NBr (TBAB) salt gave the optimal result. Changing the electric current (entry 9) led to a decrease of the yield. In addition, a combination of various electrodes was tested. On using RVC instead of a carbon rod as an anode, the yield of the desired product **3a** increased from 62% to 77% (Table 1, entry 10), probably due to the comparatively higher surface area. A Pt plate instead of the carbon rod as an anode proved to be less efficient and provided **3a** in 40% yield (entry 11), and the Ni cathode was also not as effective as the Pt plate. Finally, no product was obtained in the absence of the electric current (Table 1, entry 13) after the completion of reaction.

Encouraged by the preliminary results, we next demonstrated the generality of the present protocol by examining the scope of 2-alkynylthioanisoles **1** and sodium sulfinates **2**, as shown in Schemes 3 and 2, respectively. In general, the developed protocols have demonstrated tolerance toward various sodium sulfonates (Scheme 2), thus permitting the introduction of sulfonyl groups bearing different substituents such as halogens, CH_3, OCH_3, CF_3, and NO_2, affording the products (**3a–3h**) in medium to good yields ranging from 34% to 79%. It is noteworthy that, sodium sulfinates with an electron-withdrawing halogen (Cl) at both *para*- and *ortho*-positions gave relatively low yields (**3e** and **3i**). Pleasingly, 2-naphthyl substituted sodium sulfinate also reacted smoothly to generate **3j** in an excellent yield (86%). On the other hand, thiophene-2-sulfinate showed low efficiency in this method. However, aliphatic sodium methanesulfinate did not partake in this reaction, and this might be due to the lack of stability of its sulfone radicals.

To explore the substrate scope, we then investigated a series of substrates with different 2-alkynylthioanisoles (Schemes 3). Different substituents on the aromatic ring of 2-phenylethynyl were studied, the electronic effects were minimal and the targeted products (**3n–3af**) were generated in 42%–84% yields. Of note, heteroaromatic internal alkynes were also suitable reaction partners, delivering the cascade cyclization product in 55%–64% yields. To our delight, different functional groups, such as OMe, Me, halogen on the aromatic ring of *o*-



Scheme 2 Substrate scope of sodium sulfinates. Reaction conditions: 2-Alkynylthioanisole (**1a**, 0.20 mmol), sodium sulfinate (**2**, 0.40 mmol), constant current = 2.0 mA, n-Bu_4NBr (0.20 M), $CH_3CN : H_2O$ (1 : 1, 2 mL) in an undivided cell with one platinum electrode and one RVC electrode, rt, 10 h, isolated yield. [a] RVC/RVC instead of RVC/Pt.

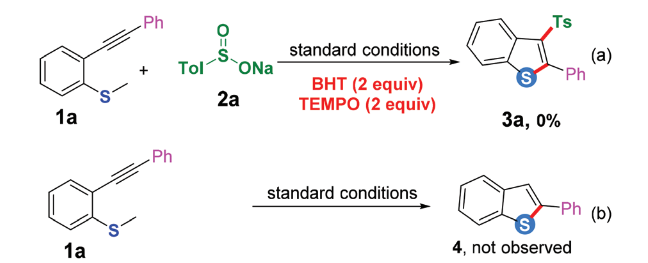


Scheme 3 Substrate scope of 2-alkynylthioanisoles. Reaction conditions: 2-Alkynylthioanisole (**1**, 0.20 mmol), sodium sulfinate (**2a**, 0.40 mmol), constant current = 2.0 mA, n-Bu_4NBr (0.20 M), $CH_3CN : H_2O$ (1 : 1, 2 mL) in an undivided cell with one platinum electrode and one RVC electrode, rt, 10 h, isolated yield.

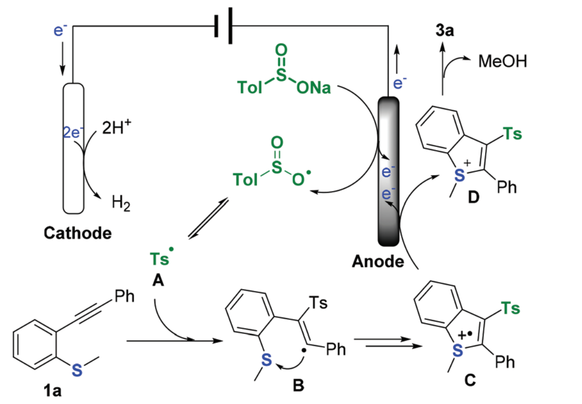
methylthio-aryl alkyne underwent the reaction smoothly and afforded the desired products in good to excellent yields. Additionally, the aliphatic internal alkyne did not couple under the optimized reaction conditions to afford desired product with the recovery of **1**. To expand the application of this electrochemical reaction, several *ortho*-substituted internal alkynes were synthesized and applied in this electrochemical cascade reaction using oxygen or nitrogen atom as a vinyl radical acceptor; however no cascade products were isolated (See Supporting Information for detail†).

To gain insight into this electrochemical cascade cyclization reaction, several mechanistic studies were conducted to understand the reaction mechanism (Scheme 4). In a radical inhibiting experiment, by adding 2 equiv. of BHT or 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) under standard conditions, this transformation was inhibited. When **1a** was treated without **2a** under standard conditions, the benzothiophene product **4** was not obtained. Next, cyclic voltammetry (CV) experiments were performed to investigate the redox potential of substrates. As shown in Fig. S1 (Supporting Information)†, a mixture of **1a** and TBAB had no obvious oxidation peak (red curve b) in the range of 0.0–2.0 V *vs.* Ag/AgCl, which revealed that compound **1a** could not be oxidized within the examined potential window. Meanwhile, TsNa readily underwent anodic oxidation, and its oxidative peak appeared at 0.95 V *vs.* Ag/AgCl (blue curve c). The same oxidative peak at 0.98 V *vs.* Ag/AgCl was observed (green curve d) when a mixture of **1a**, **2a** and TBAB was employed. These results implied that radical intermediates are possibly involved in this electrochemical oxidation system, and that TsNa was likely to be first oxidized at the anode.

Based on these control experiments and cyclic voltammetric examination, a possible mechanism is illustrated in Scheme 5. Initially, TsNa was likely to be first oxidized at the anode to generate the Ts radical **A**. Subsequently, the radical addition of **A** to the alkyne functionality in **1a** offers a vinyl radical **B.** After



Scheme 4 Control experiments.



Scheme 5 Proposed mechanism.

the vinyl radical intermediate **B** is formed, its addition to the sulfur atom *via* intramolecular cyclization/oxidation generates the radical cation **C**.[14] The soon-to-be followed oxidation of **C** proceeded smoothly to give the intermediate cation **D**, which then underwent demethylation to produce **3a** and CH_3OH with hydroxyl anions. The concomitant cathodic half-reaction involves the reduction of two protons to H_2.

In summary, an electrochemically efficient and practical radical addition/annulation cascade reaction of internal alkynes with sodium sulfinates was established at room temperature, accessing 3-sulfonated benzothiophenes under catalyst- and chemical-oxidant-free conditions. This strategy features easily accessible substrates, operational simplicity, excellent functional group tolerance and good yields. Given that the electrochemical experiment setups are widely available nowadays, the established electrochemical radical cascade of internal alkynes may be applicable to the preparation of other complex sulfur-containing heterocyclic compounds in the future. Further exploration for other electrochemical studies of sodium sulfinates or 2-alkynylthioanisoles is underway in our laboratory.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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